Pi-hole Watchdog Setup Guide

Oct 24, 2026 · @Daniel

A small script that runs on your Pi-hole and pushes an alert to your phone when Pi-hole, Unbound or the Pi's hardware has a problem, each with a suggested fix. Setup takes about 10 minutes.

What it does

The watchdog runs on the Pi itself and only messages your phone when something needs attention. Every alert ends with a **Fix:** line saying what to do.

Check	How often	What it catches
Pi-hole running and answering	Every 30 min	Pi-hole crashed or stopped answering, so the house has no internet
Ad blocking working	Every 30 min	Blocklists missing or blocking switched off
Unbound answering (if installed)	Every 30 min	Unbound broken, often by a router feature that intercepts DNS
Pi clock in sync	Every 30 min	A wrong clock, which makes Unbound fail
CPU temperature, fan	Every 30 min	Overheating, blocked vents, a dead fan (fan speed on Pi 5 only)
Power supply voltage	Every 30 min	A weak power supply or cable, the top cause of SD card corruption
SD card health and space	Every 30 min	Read/write errors, a card gone read-only, a card over 90% full
Memory, unexpected reboots	Every 30 min	Memory nearly full; restarts from power outages or crashes
Emergency guard	Every minute	85°C or failing power for 3 minutes straight: shuts the Pi down cleanly

Check	How often	What it catches
Daily summary	5:30 AM	Lookups and blocked % for the last 24 hours, plus a health line

How alerts behave:

- **Fixes itself first.** On a failure it restarts the broken service once. If that works, you get a quiet "fixed itself" notice.
- **Backup DNS.** If Unbound breaks and a restart doesn't fix it, Pi-hole switches to Cloudflare so the house stays online, then switches back by itself when Unbound recovers.
- **No false alarms.** Pi-hole problems must fail two checks in a row before you're alerted.
- **Repeats.** A problem that lasts is re-sent every 4 hours, marked "still happening", until it's fixed. Then you get one all-clear.
- **Snooze 3h button** on every alert that repeats. It works from anywhere. The emergency shutdown alert is never snoozed.
- **Offline-safe.** Alerts that can't be sent while your internet is down are held and delivered once it's back.

Before you start

You need:

- A Raspberry Pi running **Pi-hole v6** on Raspberry Pi OS (or Debian), with Pi-hole already working.
- An SSH login to the Pi (PuTTY on Windows, Terminal on Mac) with an account that can use `sudo`.
- A phone with the free **ntfy** app (Android or iPhone).
- Unbound is optional. The installer detects it on port 5335 and turns on the Unbound checks and backup only if it's there.

One-time fix so the Pi recovers after a power outage. A Pi has no clock battery, so it boots with the wrong date. If the Pi looks up names through itself, or through the router that points back at it, it can't reach a time server to fix the clock, and Unbound then rejects every answer. Check with:

```
cat /etc/resolv.conf
```

If the `nameserver` line shows `127.0.0.1`, `::1`, the Pi's own address or the router's address, point the Pi's own lookups at Cloudflare. Your other devices still go through Pi-hole. Find the connection name first:

```
nmcli -t -f NAME connection show --active
```

Then use that name (not `lo`) in place of `Wired connection 1`:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dns "1.1.1.1 1.0.0.1"
ipv4.ignore-auto-dns yes
sudo nmcli connection up "Wired connection 1"
cat /etc/resolv.conf
```

The last command should now show `nameserver 1.1.1.1`. The SSH session may pause for a few seconds during the middle command.

If your router has parental controls or a "security" add-on (for example Netgear Circle or Armor), it may intercept DNS. That breaks Unbound and can flood Pi-hole with lookups. Turn it off if you use Unbound.

Step 1: Install the ntfy app

1. Install **ntfy** from the Play Store or App Store. It's free and needs no account.
2. Don't subscribe to anything yet. The installer in Step 2 creates your private channel name.
3. Leave ntfy's **instant delivery** setting off. Normal delivery arrives within seconds and uses almost no battery.

Step 2: Run the installer

The installer comes as a separate file, **pihole-watchdog-install.sh**, sent along with this guide. Use that file rather than copying code out of a PDF or Word document, which can quietly break it.

1. Save the installer file, pihole-watchdog-install.sh, that came with this guide. Don't open and re-save it in Notepad, which can change the line endings and break it.
2. Copy it to the Pi. On Windows, open PowerShell in the folder where you saved it and run the line below, using your Pi's username and address. WinSCP works too if you'd rather drag and drop.

```
scp pihole-watchdog-install.sh pi@192.168.1.XX:~
```

3. SSH into the Pi and run it:

```
bash pihole-watchdog-install.sh
```

When it finishes, it prints a box like this one. **Write down the channel name.** It's your private alert channel.

```
================================================
  Your ntfy channel name:  pihole-k3x9q2m7w1ab
  Router: 192.168.1.1    This Pi: 192.168.1.82    Unbound: yes
================================================
```

What the installer does:

- Saves the script to `/usr/local/bin/pihole-watchdog.sh`.
- Creates your settings in `/etc/pihole-watchdog.conf`: a random channel name, plus your router address, Pi address and whether Unbound is present, all detected automatically.
- Schedules the checks every 30 minutes, the 5:30 AM summary and the every-minute emergency guard.
- Prints a NOTE if the Pi still needs the power-outage fix from Before you start.

Running it again later, for example to update, keeps your settings and channel name.

Step 3: Subscribe and test

1. In ntfy, tap **+**, type the channel name the installer printed, and subscribe.
2. Send a test alert:

```
sudo pihole-watchdog.sh test
```

You should see `Sent` and `Current checks: all passing` in SSH, and a **Pi-hole watchdog: Test message** on your phone.

See a real alert. This runs the real checks with a fake 81°C temperature. It uses a separate test folder, so the real watchdog isn't affected, and 81°C is below the 85°C shutdown point:

```
sudo rm -rf /tmp/wd-sim && echo 81000 > /tmp/fake_temp && sudo
STATE_DIR=/tmp/wd-sim THERMAL=/tmp/fake_temp /usr/local/bin/pihole-
watchdog.sh check
```

A **Pi hardware warning** arrives with a Fix line and a **Snooze 3h** button. You can run this line as often as you like. Each run starts fresh.

Try the Snooze button. Expand the notification (or open the ntfy app) and tap **Snooze 3h**. Within about a minute a **Repeat alerts snoozed** message confirms the Pi received it. Nothing is wrong right now, so the snooze has no effect. To cancel it:

```
sudo rm -f /var/lib/pihole-watchdog/snooze_until
```

Never run the `guard` command by hand with a fake temperature. That's the one that really shuts the Pi down.

To see what the watchdog has been doing:

```
sudo tail -n 20 /var/lib/pihole-watchdog/watchdog.log
```

Customizing

All settings live in one file. Open it with `sudo nano /etc/pihole-watchdog.conf`, remove the `#` in front of a line to change it, then save with Ctrl+O, Enter, Ctrl+X. Changes apply on the next check.

Setting	Default	What it does
`TEMP_WARN`	75	Warn at this CPU temperature (°C)
`GUARD_HOT`	85	Emergency shutdown at this CPU temperature (°C)
`GUARD_MINUTES`	3	Minutes in a row of overheating or failing power before shutting down
`DISK_WARN`	90	Warn when the SD card is this % full
`MEM_WARN`	10	Warn when free memory drops below this %
`REPEAT_HOURS`	4	Re-send an alert every N hours while a problem lasts

Setting	Default	What it does
`SNOOZE_HOURS`	3	How long the Snooze button silences repeats
`BACKUP_DNS`	Cloudflare (1.1.1.1, 1.0.0.1)	Where Pi-hole sends lookups if Unbound breaks
`USE_UNBOUND`	detected	1 = check Unbound and use the backup, 0 = skip

To change the 5:30 AM summary time or the 30-minute check interval, edit `/etc/cron.d/pihole-watchdog`. The first two numbers on each line are the minute and the hour.

To silence alerts for a while without changing anything, tap the channel's **bell icon** in ntfy and pick a mute duration.

Turning it off or uninstalling

Pause all checks, including the emergency guard, by removing the schedule. Run the Step 2 installer again to turn them back on.

```
sudo rm -f /etc/cron.d/pihole-watchdog
```

Remove it completely:

```
sudo rm -f /etc/cron.d/pihole-watchdog /usr/local/bin/pihole-watchdog.sh
/etc/pihole-watchdog.conf
sudo rm -rf /var/lib/pihole-watchdog
```

If the watchdog had switched Pi-hole to the backup DNS when you removed it, switch Pi-hole back to Unbound in **Settings > DNS** (custom upstream `127.0.0.1#5335`). Then delete the channel in the ntfy app.